



Security Overview

- Hackers are always on the lookout for server vulnerabilities. It is your responsibility to ensure your data is safe and secure.
- Minimize risks and be confident your data is safe on secure servers by implementing server security tips and best practices.
 - **Establish and Use a Secure Connection**
 - **Use SSH Keys Authentication**
 - **Secure File Transfer Protocol(SFTP)**
 - **Secure Sockets Layer Certificates(SSL)**
 - **Use Private Networks and VPNs**
 - **Monitor Login Attempts**
 - **Manage Users**
 - **Establish Password Requirements**

Security Overview

- **Set Password Expiration Policy**
- **Use Passphrases For Server Passwords**
- **Do not use the same password for multiple accounts**
- **Update and Upgrade Software Regularly**
- **Remove or Turn Off All Unnecessary Services**
- **Hide Server Information**
- **Use Intrusion Detection Systems**
- **File Auditing**
- **Service Auditing**
- **Set Up and Maintain a Firewall**
- **Backup Your Server**
- **Create Multi-Server Environments**
- **Create Virtual Isolated Environments**

Security Overview

- **Set File permissions**
- **Encrypting sensitive data**
- **Regularly monitoring system logs**
- **Disable root login**
- **Disable IP forwarding**
- **Implement multi-factor authentication**
- **Use SELinux or AppArmor to enforce mandatory access controls**
- **Educate users about security best practices**

Linux Useful Command

- `$ sudo apt update -y && sudo apt upgrade -y # update os`
- `$ sudo ufw status # Enable Firewall`
- `$ sudo ufw enable`
- `$ sudo ufw allow ssh # sudo ufw allow 22/tcp`
- `$ sudo ufw allow 1:1000/tcp`
- `$ sudo ufw reload`
- `$ chmod 700 sensitive_directory # Set File Permission`
- `$ chmod 600 sensitive_file`
- `$ sudo systemctl stop servicename.service # Stop unnecessary services`
`# sudo service servicename stop`

Linux Useful Command

- `$ sudo nano /etc/ssh/sshd_config # Disable root login`
`PermitRootLogin no`
- `$ sudo systemctl restart ssh`
- `$ sudo passwd username # Change user password`
- `$ tar -cvf file.tar ./ # Compress file`
- `$ tar -xvf file.tar # Extract file`
- `$ netstat -tulpn # Check Services and Ports`
- `$ sudo apt install net-tools`
- `$ ifconfig # Check ip and network interface`
- `$ nslookup google.com or 192.167.0.1 # Get info of domain or ip`

Linux Useful Command

- `$ telnet 192.167.0.1 8081 # Test Connection`
- `$ ping google.com # Test Connection`
- `$ route -n # Show IP Gateway`
- `$ traceroute google.com # Check traffic of domain`
- `$ curl google.com # Visit website`
- `$ wget google.com # Download file`
- `$ host google.com or host 74.125.24.102 # Show All IP of domain`
- `$ df -h / $ du -shc * # Check disk space`
- `$ free -h # Check Memory`
- `$ htop # Check CPU and Memory`

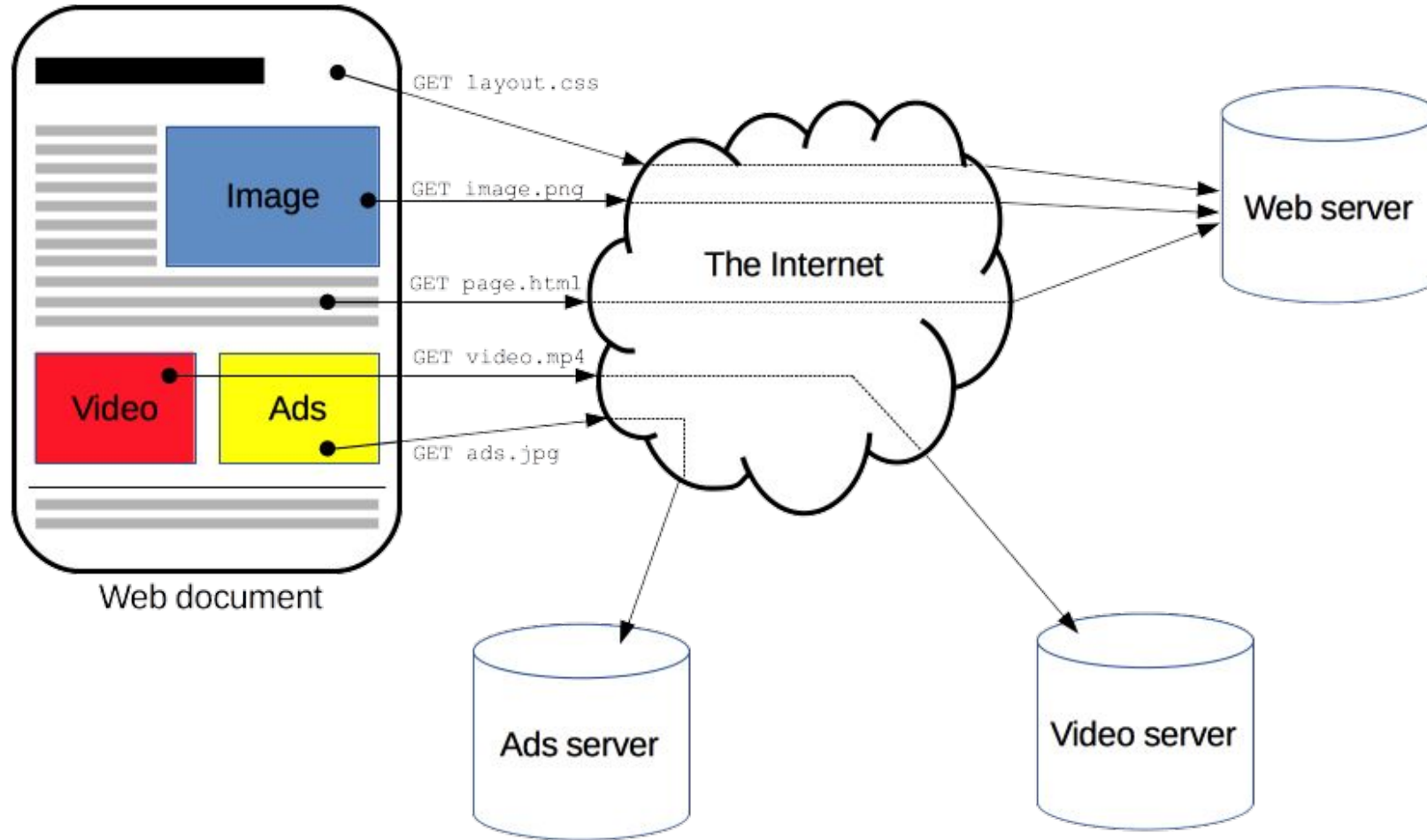
Linux Useful Command

- `$ ssh -i ~/.ssh/id_rsa.pub dara@3.0.102.244 # Login to server with ssh`
- `$ ssh dara@3.0.102.244 # Login to server with username & password`
- `$ scp dara@3.0.102.244:/home/dara/while.sh ./ # Copy file from server`
- `$ scp -r db.sql dara@3.0.102.244:/home/dara # Copy file to server`

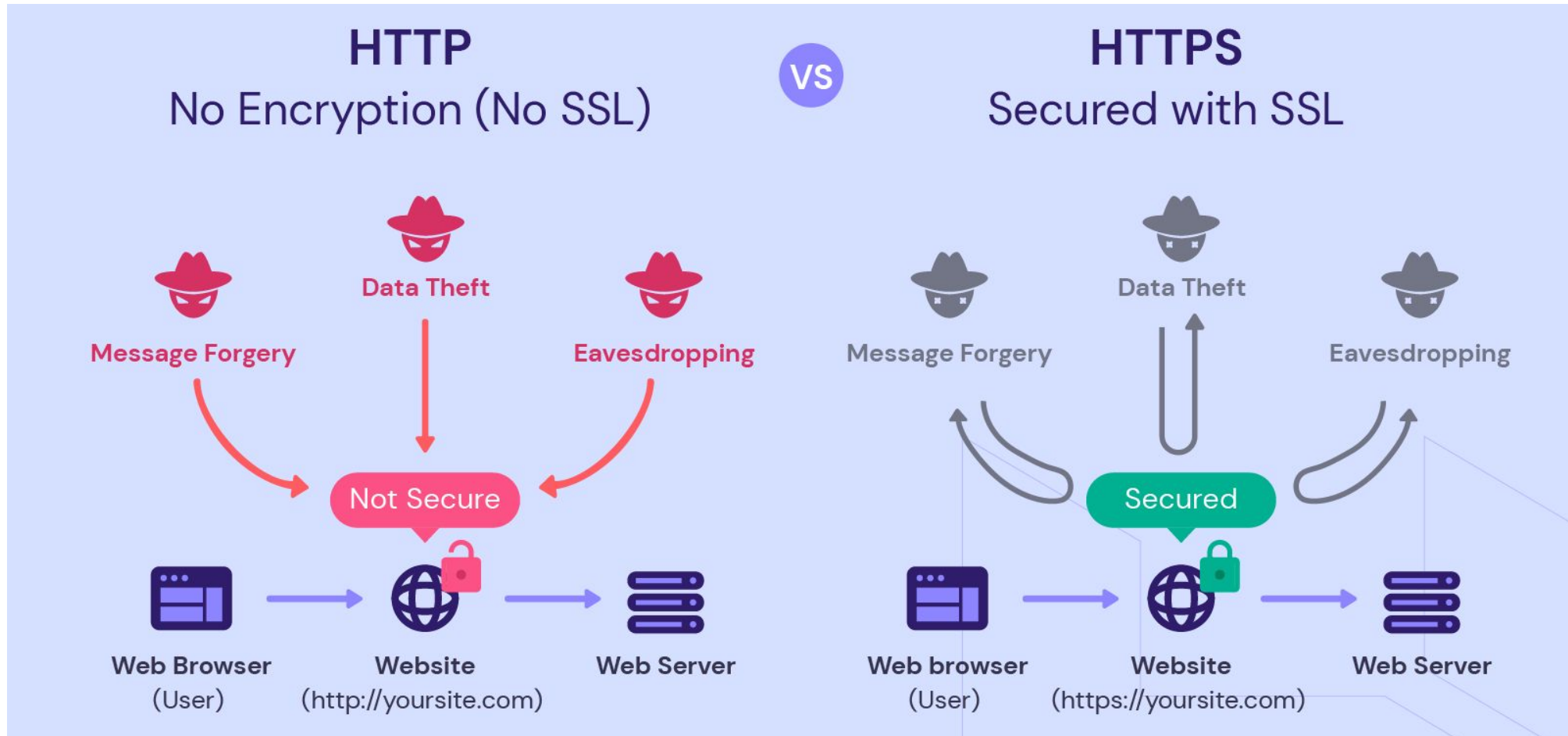
What is HTTP?

- HTTP stands for HyperText Transfer Protocol
- **HTTP** is a protocol for fetching resources such as HTML documents. It is the foundation of any data exchange on the Web and it is a client-server protocol, which means requests are initiated by the recipient, usually the Web browser. A complete document is reconstructed from the different sub-documents fetched, for instance, text, layout description, images, videos, scripts, and more.
- Communication between client computers and web servers is done by sending HTTP Requests and receiving HTTP Responses
- **Clients** are often browsers (Chrome, Edge, Safari), but they can be any type of program or device and **Servers** are most often computers in the cloud.

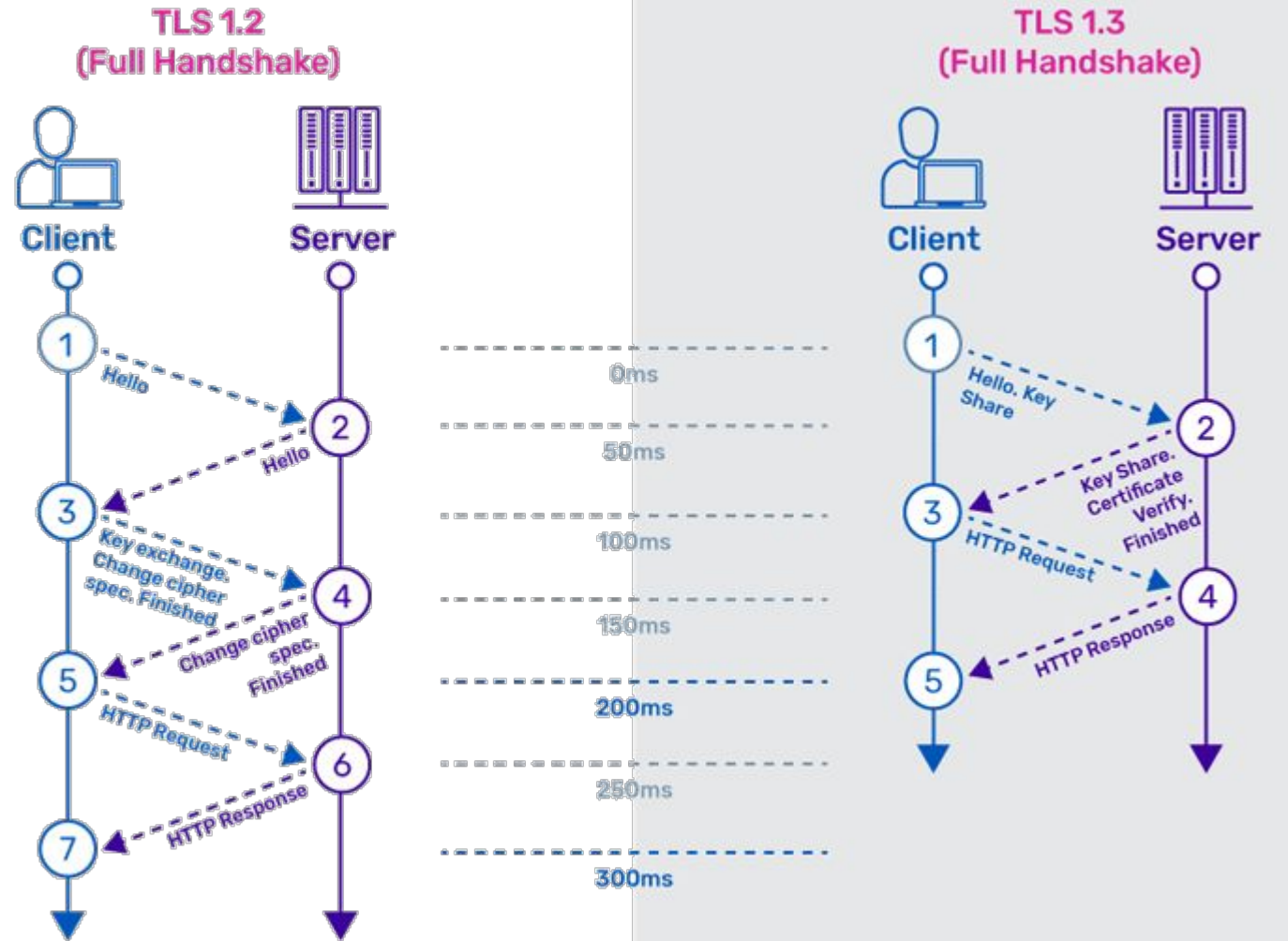
What is HTTP?



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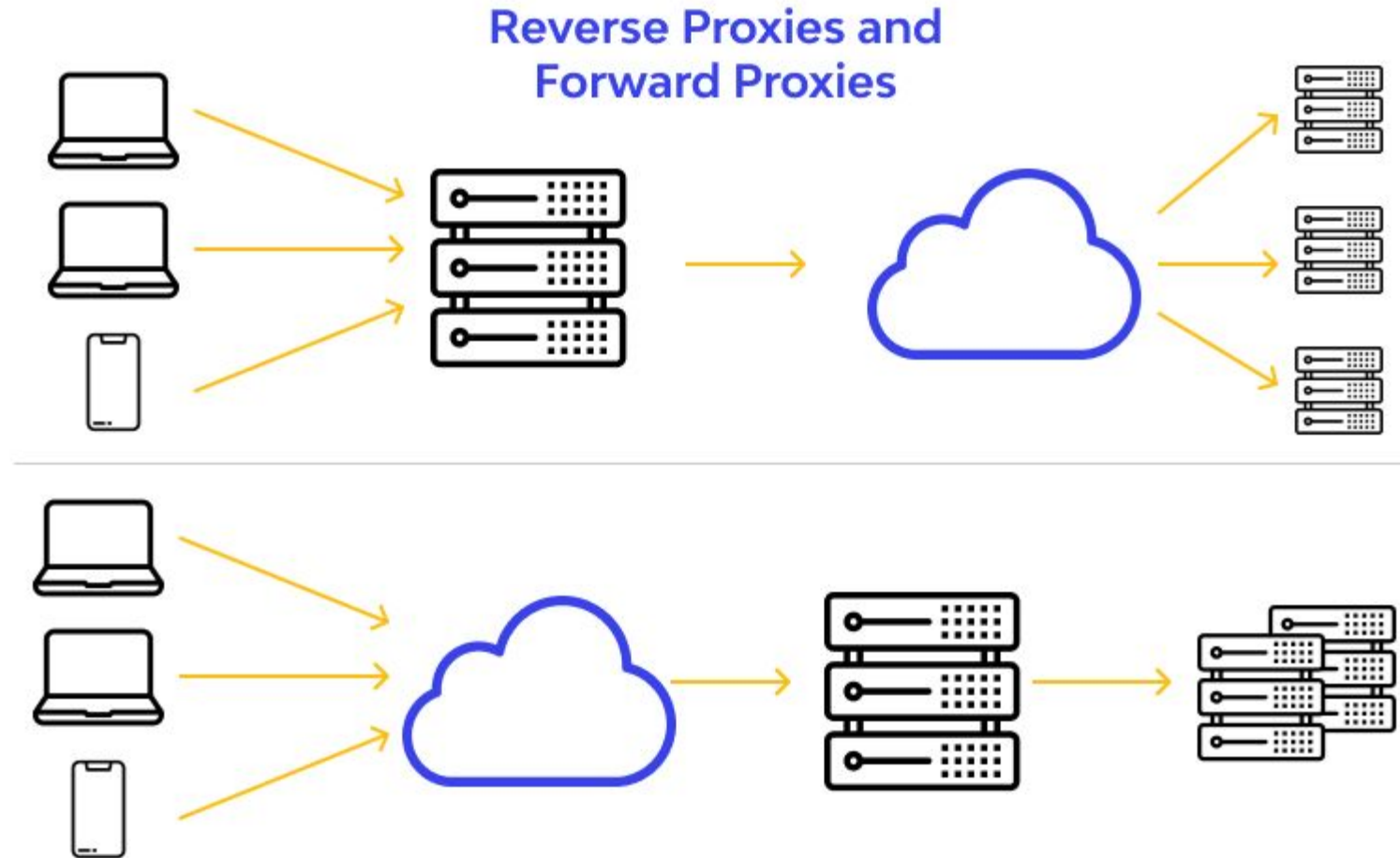
TLS/SSL Handshake



What is Proxy Server?

- A proxy server is an intermediate program or computer used when navigating through different networks of the Internet. They facilitate access to content on the World Wide Web. A proxy intercepts requests and serves back responses; it may forward the requests, or not (for example in the case of a cache), and it may modify it (for example changing its headers, at the boundary between two networks).
- There are two main types of proxy servers:
 - A **forward proxy** that handles requests from and to anywhere on the Internet.
 - A **reverse proxy** taking requests from the Internet and forwarding them to servers in an internal network.

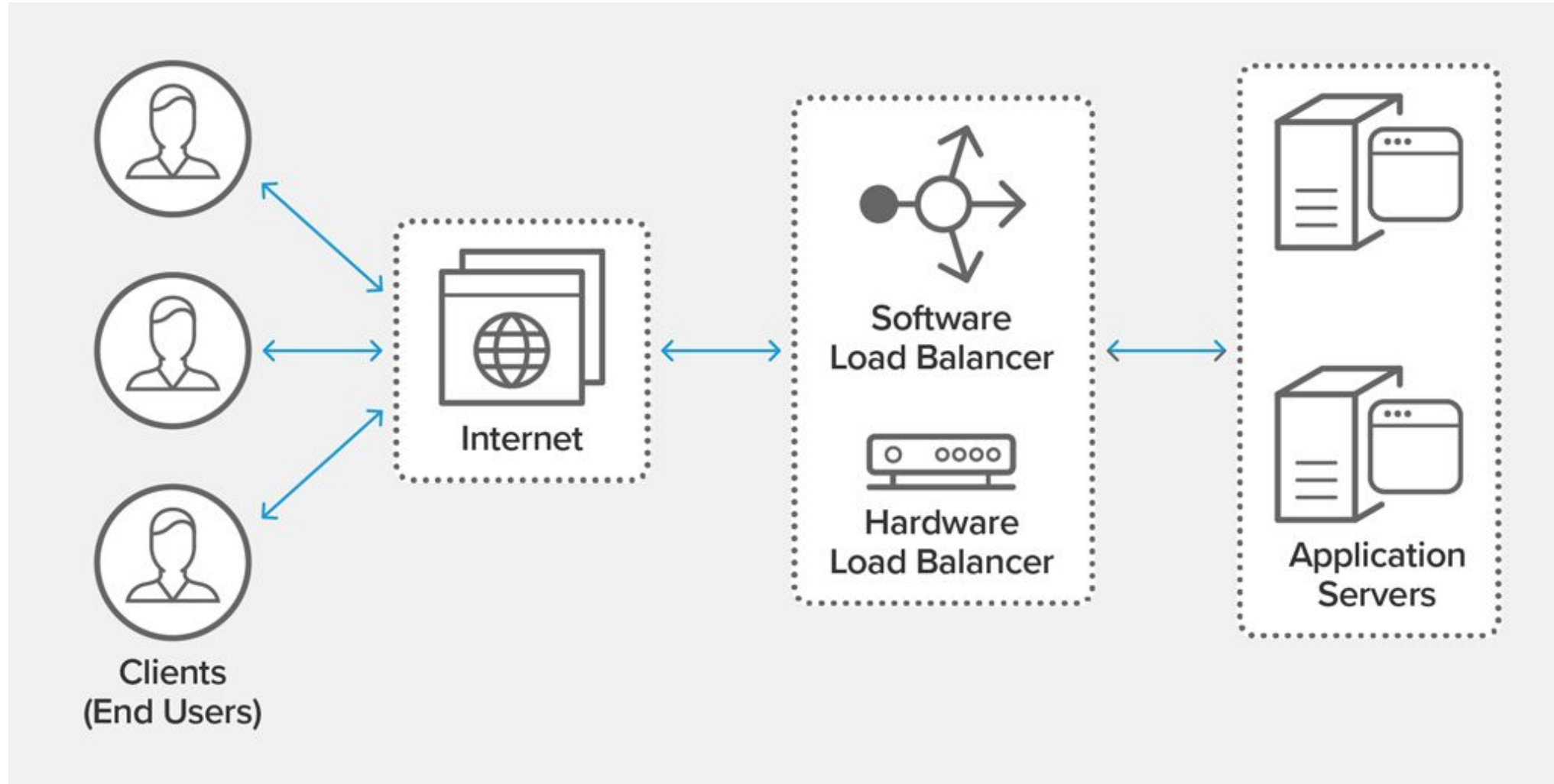
What is Proxy Server?



Load Balancer

- A load balancer acts as the “traffic cop” sitting in front of your servers and routing client requests across all servers capable of fulfilling those requests in a manner that maximizes speed and capacity utilization and ensures that no one server is overworked, which could degrade performance. If a single server goes down, the load balancer redirects traffic to the remaining online servers. When a new server is added to the server group, the load balancer automatically starts to send requests to it.

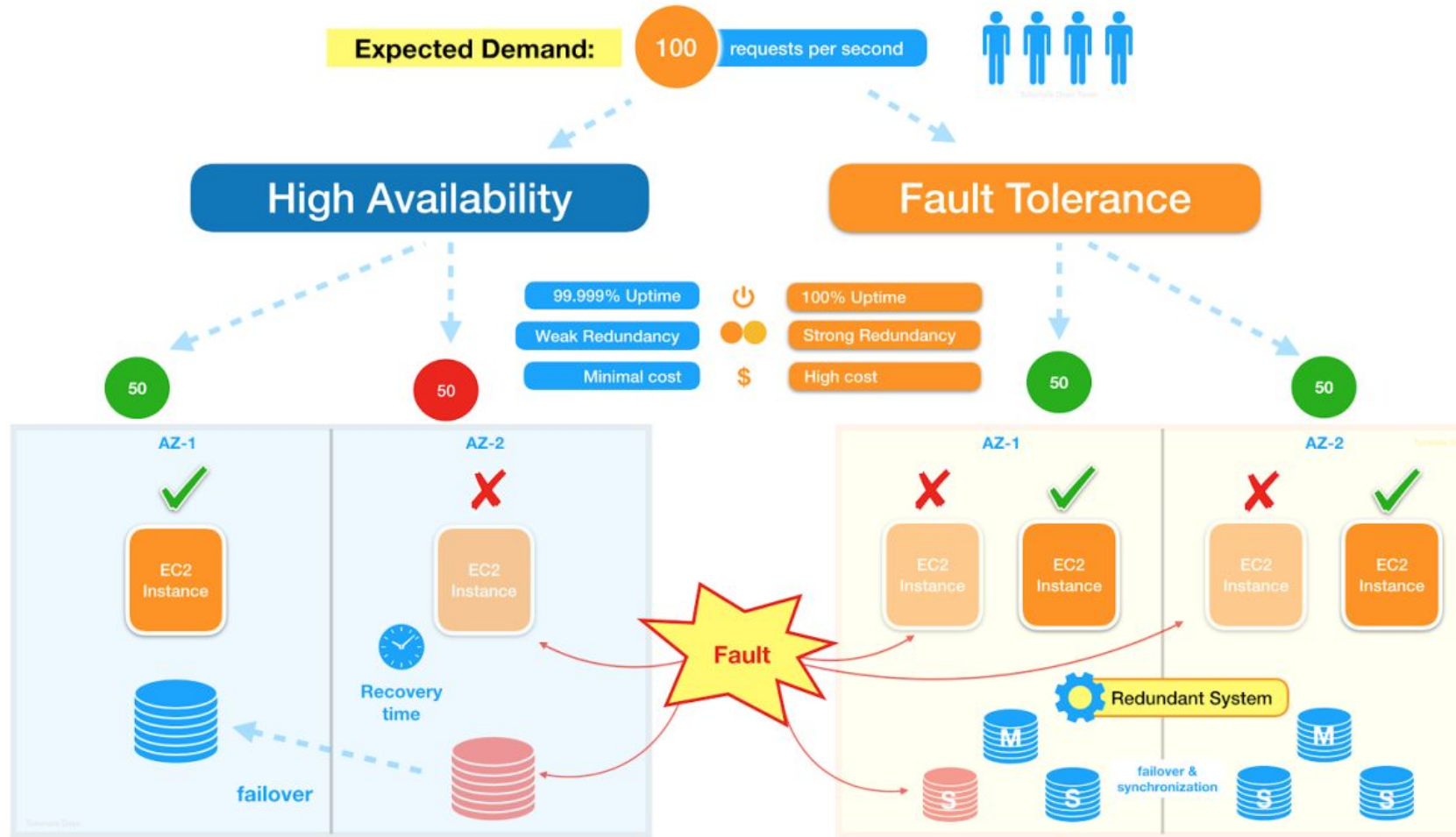
Load Balancer



High Availability vs Fault Tolerance

- A High Availability is one that is designed to be available 99.999% of the time, or as close to it as possible. Usually this means configuring a failover system that can handle the same workloads as the primary system.
- A Fault Tolerant is extremely similar to HA, but goes one step further by guaranteeing zero downtime. HA still comes with a small portion of downtime, hence the ideal of a perfect HA strategy reaching “five nines” rather than 100% uptime.

High Availability vs Fault Tolerance



Tutorials Dojo

A large, faint watermark of the ISTAD logo is centered in the background. It is a circular emblem with a blue outer ring containing the text 'INSTITUTE OF SCIENCE AND TECHNOLOGY ADVANCED DEVELOPMENT' and Khmer text. Inside the ring is a blue circle with a white atomic symbol. The word 'ISTAD' is written in red across the bottom of the inner circle. The text 'Thank you' is overlaid in the center in a bold, dark blue font.

Thank you

Begin with the theory, Start with the practical